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ORIGINAL CONTRIBUTION

Potential Missed Opportunities to Administer Intravenous Thrombolysis to Patients With Acute Ischemic Stroke

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BACKGROUND: Recent years have seen improvements in stroke-care pathways, including mobile stroke units (MSUs), and this may have affected the rate, predictors, and outcomes of patients with acute ischemic stroke who qualify for but do not receive treatment with intravenous thrombolysis (IVT).

METHODS: This was a secondary observational cohort analysis of the prospective, multicenter BEST-MSU trial (Benefits of Stroke Treatment Delivered by a Mobile Stroke Unit Compared With Standard Management by Emergency Medical Services), conducted in 7 US cities from 2014 to 2020, comparing MSU management versus standard emergency department management for patients with suspected acute ischemic stroke. The analytical cohort comprised enrolled patients with confirmed acute ischemic stroke and no guideline contraindications to IVT. The outcome was a potential missed IVT opportunity, defined as patients not treated with IVT despite lacking contraindications. We used multivariable logistic regression to evaluate whether demographics, prestroke modified Rankin Scale, study site, comorbidities, National Institutes of Health Stroke Scale, blood pressure, international normalized ratio, glucose, antithrombotic use, and thrombectomy were associated with this outcome.

RESULTS: Of 1515 enrolled patients, 927 met criteria for this analysis. Fifty-one participants (5.5%) had a potential missed IVT opportunity, including 4 of 555 (0.7%) in the MSU group versus 47 of 372 (12.6%) in the emergency department group (odds ratio, 0.05 [95% CI, 0.02–0.12]). In multivariable analysis, omitting study group, lower National Institutes of Health Stroke Scale (odds ratio, 0.95 [95% CI, 0.90–0.99]) and longer last known well-to-door time (odds ratio per 10 minutes, 1.08 [95% CI, 1.03–1.13]) were independently associated with a potential missed IVT opportunity. The leading reasons documented for withholding IVT were resolving symptoms (43%), time window concerns (18%), and minor deficits (10%). Among participants with a potential missed IVT opportunity and a recorded 3-month modified Rankin Scale (n=49), 19 (39%) had a score of 3 to 6.

CONCLUSIONS: In the BEST-MSU trial, potential missed IVT opportunities occurred in 1-in-8 patients in the emergency department, and rarely on the MSU. Although often due to early improvement or minor deficits, more than one-third of these patients had poor functional status at 3 months.

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Key Words: blood pressure ■ functional status ■ ischemic stroke ■ odds ratio ■ thrombectomy

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Nonstandard Abbreviations and Acronyms

AIS	acute ischemic stroke
aOR	adjusted odds ratio
ARAMIS	Antiplatelet vs R-tPA for Acute Mild Ischemic Stroke
ED	emergency department
IQR	interquartile range
IVT	intravenous thrombolysis
LKW	last known well
mRS	modified Rankin Scale
MSU	mobile stroke unit
NIHSS	National Institutes of Health Stroke Scale
OR	odds ratio

Stroke is the fourth leading cause of death and a leading cause of adult disability in the United States.[1,2] Intravenous thrombolysis (IVT) is the cornerstone of acute ischemic stroke (AIS) treatment.[3] Randomized trials have shown that IVT administered within 4.5 hours of onset greatly improves the odds of an excellent neurological outcome.[4] However, prior studies have reported that up to 40% of patients with AIS arriving at the hospital during this timeframe do not receive IVT even in the absence of contraindications.[5–7] The leading reasons for withholding IVT in these studies were mild or resolving symptoms, although previous reports suggest that ≈30% of such patients can't ambulate independently at discharge and 10% to 20% are disabled at 3 months.[5–9]

Over the past 20 years, through national quality improvement initiatives, public awareness campaigns, prehospital transport protocols, imaging advancements, stroke center certifications, organized stroke teams and alerts, better secondary prevention, and enhanced rehabilitation programs, there have been considerable improvements in stroke quality of care and outcomes in the United States.[10] Additionally, some regions have access to mobile stroke units (MSUs), an idealized stroke-care system proven to increase and expedite IVT for AIS.[11,12] These various interventions could have affected the rate, predictors, and outcomes of patients with AIS not treated with IVT; however, there is a dearth of modern prospective data on this topic. To address this knowledge gap, we performed a secondary analysis of a contemporary prospective trial comparing MSU transport and treatment to standard ambulance transport and emergency department (ED) treatment for patients with suspected AIS presenting within 4.5 hours of onset.[13] Our prespecified hypotheses were that (1) potential missed opportunities to administer IVT for AIS occur in <5% of patients treated aboard a MSU, a rate significantly lower than in the ED, (2) potential missed IVT opportunities are more common in patients with minor deficits and baseline disability regardless of setting, and (3) patients with missed IVT opportunities have worse clinical outcomes than similar patients who receive IVT.

METHODS

Design and Setting

This was a secondary observational cohort analysis of existing data from the BEST-MSU trial (Benefits of Stroke Treatment Delivered by a Mobile Stroke Unit Compared with Standard Management by Emergency Medical Services). BEST-MSU was a prospective, multicenter, controlled trial conducted in 7 US cities (Houston, Colorado, Memphis, New York, Los Angeles, Burlingame, and Indianapolis) from 2014 to 2020. Through an alternating-week design, MSU management was compared with standard emergency medical services and ED management for patients with suspected AIS meeting initial screening criteria for IVT treatment.[13] All participating study sites were comprehensive stroke centers with around-the-clock stroke teams trained and experienced in administering IVT in the ED. Details of the trial's design and protocol have been previously published.[13] The institutional review boards of all participating sites approved the study, and written informed consent was obtained from all participants or their proxies. This article follows the STROBE reporting guideline.[14] The data from this analysis is available by application to the trial's governing board.

Population

Patients were eligible for enrollment into the BEST-MSU trial if, at the time of prehospital presentation to the MSU or standard ambulance paramedic team, they had a history and physical exam consistent with AIS, were within 4.5 hours of last known well (LKW) time, and met standard guideline-based criteria for IVT before a head computed tomography scan or laboratory evaluation was performed.[15] The analytical cohort for the trial's primary analysis was patients ultimately adjudicated as having been IVT eligible on presentation (ie, lacking contraindications), as determined by an expert neurologist incorporating computed tomography and laboratory data while blinded to treatment group. For the current analysis, as our focus was on potential missed IVT opportunities, we included trial participants who had a final adjudicated diagnosis of AIS, were evaluated by the MSU or arrived at the ED within 4.5 hours of LKW, and had no absolute guideline contraindications to thrombolysis according to the expert adjudicator.[15]

Measurements

Study exposures included group assignment, age, sex, race, ethnicity, baseline modified Rankin Scale (mRS) score, study site, prior antithrombotic use (defined as any baseline antiplatelet or anticoagulant medications), initial National Institutes of Health Stroke Scale (NIHSS) score, time from LKW to door (on scene time for MSU patients), first recorded systolic blood pressure, first recorded finger stick glucose, first recorded international normalized ratio, performance of endovascular therapy, and the following comorbidities: history of diabetes, heart failure, coronary artery disease, hypertension, and hyperlipidemia.

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We also collected patient outcome data through 3-month mRS scores. All these measurements were collected prospectively by trained study investigators.

The study outcome was a potential missed opportunity to administer IVT for AIS. This was defined as patients with a final diagnosis of AIS, who were evaluated by the MSU or arrived at the ED within 4.5 hours of LKW and had no absolute guideline contraindications to IVT according to the expert adjudicator, but did not receive IVT treatment. To investigate reasons why eligible patients were not treated with IVT, study investigators performed a post hoc review of comprehensive electronic health records and study forms. Importantly, these study forms had structured sections on IVT eligibility and contraindications, which were filled out separately by local site investigators and an expert adjudicator blinded to the study arm. Documented reasons for not treating are reported.

Analysis

Descriptive statistics were used to characterize participants' baseline characteristics and clinical outcomes. Continuous variables were reported as mean (SD) or median (interquartile range [IQR]) depending on their data distribution, while categorical variables were reported as counts (percentages). Fischer exact test was used to compare the proportion of potential missed IVT opportunities (study outcome) between the MSU and standard ED management groups. The distribution of documented reasons for withholding IVT was calculated according to treatment group. Univariate and multivariable logistic regression were used to evaluate the relationships between covariates and the outcome of a potential missed IVT opportunity (no IVT treatment versus IVT treatment in eligible patients). LASSO (Least Absolute Shrinkage and Selection Operator) regularization and clinical rationale were used to build the final multivariable model. Age, sex, race, ethnicity, baseline mRS, initial NIHSS score, and time from LKW-to-door were forced into the model regardless of statistical significance at the univariate level. The study group (MSU versus standard ED management) was excluded from the multivariable model so the findings would generalize to all acute stroke treatment settings.

Subgroup analyses were performed among patients with an NIHSS of ≥4 (ie, patients with moderate or severe stroke severity) and those in the standard management group.

In exploratory analysis, we report the 3-month mRS scores among participants in the primary analytical cohort according to whether they received IVT. We also compared the baseline and 3-month mRS scores among participants who were not treated with IVT. Further, to investigate the effects of withholding IVT on functional outcomes, we used propensity scores, via the nearest neighbor approach, to match participants not treated with IVT to participants treated with IVT in a 1:2 ratio by age, sex, baseline mRS, initial NIHSS, and LKW-to-door time. We used multivariable logistic regression to compare rates of good 3-month functional outcomes (mRS score, 0–2) between these groups.

In secondary analysis, we evaluated participants who had a final adjudicated diagnosis of AIS, were evaluated by MSU or arrived at the ED within 4.5 hours of LKW, and had no absolute contraindications to IVT according to either the expert adjudicator or the primary treating team (MSU physician for the MSU group and emergency medicine or neurology physicians for the standard management group). Using this broader secondary cohort, which included patients who a primary treating physician(s) thought was eligible for IVT, but the expert adjudicator did not, we listed the number and reasons for potential missed IVT opportunities by study group.

The analyses were performed by M.W. and J.-M.Y. using R software. Missing data were rare and thus not imputed. Statistical significance was set at an alpha error of 0.05.

RESULTS

Participant Characteristics

Of 1515 patients enrolled in the master trial, 927 had a confirmed AIS and were eligible for IVT, meeting criteria for this analysis. A study flow diagram is provided in Figure 1. Participants' median age was 67.0 (IQR, 57.0–79.0) years, and 477 (51.5%) were men (Table 1). Their race-ethnicity was self-reported as 39.7% non-Hispanic Black, 36.7% non-Hispanic White, 17.0% Hispanic, and 6.6% other. Baseline disability, defined as an mRS score of 3 to 5, was present in 155 participants (16.7%). Most participants were enrolled in Houston (n=722, 77.9%). Comorbidities were common, particularly hypertension (77.9%) and hyperlipidemia (42.4%). The median (IQR) baseline NIHSS score was 9 (6–16). Median (IQR) time from LKW to door was 54 (37–88) minutes. Endovascular therapy was performed in 262 (28.3%) participants.

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Potential Missed Opportunities Rate and Documented Explanations

Fifty-one participants (5.5%) had a potential missed IVT opportunity, including 4 of 555 (0.7%) in the MSU group versus 47 of 372 (12.6%) in the standard management group (odds ratio [OR], 0.05 [95% CI, 0.02–0.12]). Table 2 lists the reasons documented by treating clinicians for withholding IVT. The leading reasons were resolving symptoms (n=22, 43.1%), time window concerns (n=9, 17.6%), minor deficits (n=5, 9.8%), patient/surrogate declined (n=4, 7.8%), and history of intracranial hemorrhage (n=3, 5.9%).

Associated Factors

In univariate analysis, aside from treatment group as described above, factors associated with a potential missed IVT opportunity included a lower baseline NIHSS (OR, 0.95 [[95% CI, 0.91–0.99]), longer LKW-to-door time (OR per 10 minutes, 1.09 [95% CI, 1.04–1.13]), and no endovascular therapy (OR, 0.46 [95% CI, 0.20–0.93]). In multivariable analysis, as described in Table 3, associated factors were lower baseline NIHSS (adjusted odds ratio [aOR], 0.95 [95% CI, 0.90–0.99]) and longer

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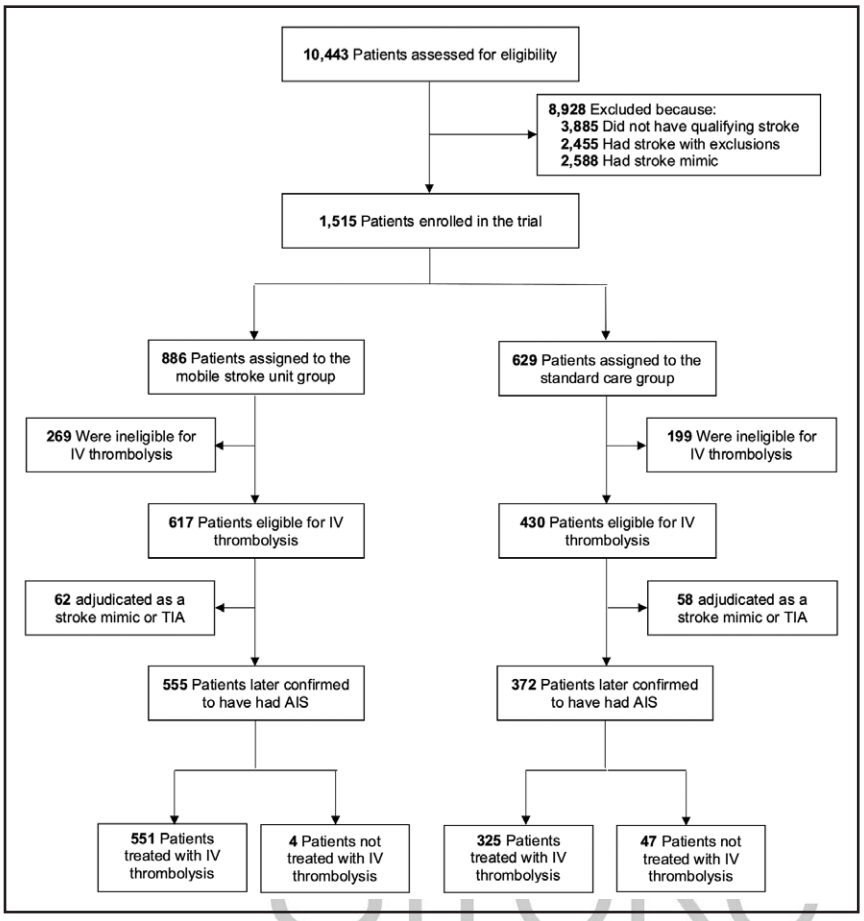

Figure 1. Study flow diagram. CONSORT flow diagram (Consolidated Standards of Reporting Trials) displaying the number of patients assessed for BEST-MSU trial (Benefits of Stroke Treatment Delivered by a Mobile Stroke Unit Compared With Standard Management by Emergency Medical Services) eligibility, the number enrolled into the trial, the number assigned to the mobile stroke unit group and the standard management group, the number eligible for intravenous (IV) thrombolysis according to study group, the number later confirmed by an expert adjudicator to have had an acute ischemic stroke (AIS), and the numbers in whom IV thrombolysis was administered or withheld according to study group. TIA indicates transient ischemic attack.

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LKW-to-door time (aOR per 10 minutes, 1.08 [95% CI, 1.04–1.13]).

Subgroup Analyses

Among participants with a baseline NIHSS ≥4, the results mirrored the primary analysis, with a lower baseline NIHSS (aOR, 0.94 [95% CI, 0.89–0.99]) and a longer LKW-to-door time (aOR per 10 minutes, 1.08 [95% CI, 1.03–1.13]) being independently associated with a potential missed IVT opportunity. When restricted to participants in the standard management group, older age (aOR, 1.03 [95% CI, 1.01–1.06]), lower baseline NIHSS (aOR, 0.946 [95% CI, 0.895–0.995]), and a longer LKW-to-door time (aOR per 10 minutes, 1.08 [95% CI, 1.02–1.14]) were independently associated with a potential missed IVT opportunity.

Exploratory Analyses: Clinical Outcomes

Figure 2 shows the 3-month mRS score distributions for participants according to whether they received IVT. Among the 51 participants with a potential missed IVT opportunity, 49 had 3-month mRS scores recorded. Their median (IQR) score was 2 (0–4), and 19 (38.8%) had a score of 3 to 6, indicating disability or death. Additionally, 27 (55.1%) participants had a worse 3-month mRS score than their baseline. Among the 27 participants who had IVT withheld because of resolving symptoms or minor deficits, 25 had 3-month mRS scores recorded. In this subgroup, 6 (24%) participants had a 3-month mRS score of 3 to 6, and 10 (40%) had a worse score than their baseline.

When participants treated with IVT were matched 2:1 by prognostic factors to participants not treated with IVT, in multivariable logistic regression analysis, rates of good functional outcomes (mRS score, 0–2) at 3 months were similar between groups (aOR, 1.23 [95% CI, 0.55–2.74]; P=0.61).

Secondary Analyses

When including the additional patients who a primary treating physician(s) thought were eligible for IVT but the expert adjudicator did not (n=980), there were a total of 62 (6.3%) participants with a potential missed IVT opportunity, including 5 of 582 (0.9%) in the MSU group versus 57 of 398 (14.3%) in the standard management group. The leading reasons for withholding IVT were resolving symptoms (n=23, 37.1%), minor deficits (n=11, 17.7%), and time window concerns (n=9, 14.5%).

Table 1. Characteristics of Patients Stratified by Administration of Intravenous Thrombolysis

Characteristic*	Total (N=927)	No IVT treatment (n=51)	IVT treatment (n=876)
Demographics			
Age, y	67 (57–79)	72 (59–86)	67 (57–79)
Male sex	477 (51.5)	25 (49.0)	452 (51.6)
Race-ethnicity			
Hispanic	158 (17.0)	12 (23.5)	146 (16.7)
Non-Hispanic Black	368 (39.7)	16 (31.4)	352 (40.2)
Non-Hispanic Other	61 (6.6)	5 (9.8)	56 (6.4)
Non-Hispanic White	340 (36.7)	18 (35.3)	322 (36.8)
Vascular risk factors			
Prior stroke or TIA	273 (29.4)	15 (29.4)	258 (29.5)
Diabetes	300 (32.4)	16 (31.4)	284 (32.4)
Heart failure	116 (12.5)	7 (13.7)	109 (12.4)
Coronary artery disease	156 (16.8)	10 (19.6)	146 (16.7)
Hypertension	722 (77.9)	37 (72.5)	685 (78.2)
Hyperlipidemia	393 (42.4)	21 (41.2)	372 (42.5)
Prior antithrombotic use	360 (38.8)	24 (47.1)	336 (38.4)
Study site			
Houston	722 (77.9)	42 (82.4)	680 (77.6)
Other sites	205 (22.1)	9 (17.6)	196 (22.4)
Stroke severity/functional status			
Initial NIHSS score	9 (6–16)	7 (5–13)	9 (6–17)
Prestroke modified Rankin Scale score	0 (0–1)	0 (0–2)	0 (0–1)
Clinical data			
Initial systolic blood pressure	160 (139–184)	157 (137–180)	160 (140–184)
Initial finger stick glucose	118 (99–152)	119 (100–152)	118 (99–152)
INR	1.0 (1.0–1.2)	1.1 (1.0–1.3)	1.0 (1.0–1.2)
Minutes from LKW to the door	54 (37–88)	75 (47–137)	53 (36–84)
Endovascular therapy	262 (28.3)	8 (15.7)	254 (29.0)
Study treatment arm			
Mobile stroke unit	555 (59.9)	4 (7.8)	551 (62.9)
Standard management	372 (40.1)	47 (92.2)	325 (37.1)

INR indicates international normalized ratio; IVT, intravenous thrombolysis; LKW, last known well; NIHSS, National Institutes of Health Stroke Scale; and TIA, transient ischemic attack.

*Data reported as n (%) for categorical variables and median (interquartile range) for continuous and ordinal variables. Percentages are rounded and may add up to >100%.

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DISCUSSION

In a contemporary, prospective, multicenter cohort of patients with AIS meeting standard eligibility criteria for IVT, ≈12% of patients in the ED and <1% of patients aboard a MSU did not receive IVT. Lower NIHSS and longer LKW-to-door time were independently associated with potential missed IVT opportunities, including among patients with an initial NIHSS of at least 4 (ie, moderate or severe stroke severity) and those treated in the ED. The leading reasons posited by treating practitioners for withholding IVT were resolving symptoms, time window concerns, and minor deficits, with resolving symptoms and minor deficits accounting for >50% of cases. Despite the decision to withhold therapy in these patients, more than one-third were disabled or dead at 3 months, and half had worse functional status than before their stroke.

Rates and predictors of potential missed IVT opportunities for AIS vary across the literature according to definitions used, location, and era.[5–7,16,17] In an analysis of hospitals participating in the American Heart Association's Get With The Guidelines-Stroke registry from 2003 to 2011, about one-quarter of 61 698 eligible patients arriving within 2 hours of AIS onset did not receive IVT, although failure rates greatly decreased during the study period, declining from 55% in 2003-2005 to 18% in 2010-2011.[6] In this large-scale analysis, older

Table 2. Documented Reasons for Withholding Intravenous Thrombolysis, Stratified by Study Group

Reason*	Overall (N=51)	Standard management (n=47)	MSU (n=4)
Resolving symptoms	22 (43.1)	20 (42.6)	2 (50.0)
Time window concerns	9 (17.6)	9 (19.1)	0 (0)
Minor deficits	5 (9.8)	5 (10.6)	0 (0)
Patient and family declined	4 (7.8)	4 (8.5)	0 (0)
History of intracranial hemorrhage	3 (5.9)	2 (4.3)	1 (25.0)
Concern for seizures	2 (3.9)	1 (2.1)	1 (25.0)
Early ischemic signs on CT	2 (3.9)	2 (4.3)	0 (0)
Anticoagulant use	1 (2.0)	1 (2.1)	0 (0)
Cerebral aneurysm	1 (2.0)	1 (2.1)	0 (0)
Direct thrombectomy preferred	1 (2.0)	1 (2.1)	0 (0)
Recent myocardial infarction	1 (2.0)	1 (2.1)	0 (0)

CT indicates computed tomogram; and MSU, mobile stroke unit.

*Documented reasons reported as n (%). Percentages are rounded and may add up to >100%.

age, female sex, non-White race, diabetes, prior stroke, atrial fibrillation, prosthetic heart valve, lower NIHSS, arrival during off-hours and not via ambulance, longer onset-to-arrival time, earlier calendar year, rural location, and nonteaching and nonstroke center hospitals were associated with potential missed IVT opportunities.[6] In a more recent analysis from 2015 to 2017 of the largest emergency care hospital in Norway, among 535 patients presenting with AIS within 4.5 hours of LKW time and no contraindications to IVT, 185 (35%) did not receive IVT.[5] The most commonly cited reasons for withholding IVT were mild or improving symptoms (37%), late arrival (13%), and advanced age (7%).[5]

In comparison to these historical data, in our study, only 5.5% of eligible AIS patients did not receive IVT, although rates were higher in the ED (12.6%) than aboard an MSU (0.7%). We also found that lower NIHSS, longer LKW-to-arrival time, and older age (only for the ED group) were associated with withholding IVT in otherwise eligible patients; and that resolving symptoms, time window concerns, and minor deficits were the most commonly posited reasons for not administering the drug. Our study differs from prior ones on this topic by analyzing a more contemporary study period, nesting within an AIS clinical trial infrastructure conducted at experienced stroke centers, and including patients treated aboard MSUs. Given the clinical trial design and considering that the primary goal of MSUs is to increase access and speed of IVT for AIS, our findings likely represent or are close to the best case scenario when it comes to stroke systems-of-care and aggressiveness of AIS treatment. That said, our data suggest that ongoing efforts to improve ED management of AIS have been associated with a continued decline in missed IVT opportunities.

Table 3. Adjusted Associations Between Clinical Factors and Potential Missed Intravenous Thrombolysis Opportunities

Factor*†	Multivariable OR (95% CI)	Multivariable *P* value
Baseline NIHSS score‡	0.95 (0.90–0.99)‡	0.033‡
LKW-to-door time, 10-min‡	1.08 (1.03–1.13)‡	<0.001‡
Age, y	1.020 (0.997–1.045)	0.092
Male sex	1.01 (0.56–1.81)	0.978
Hispanic race-ethnicity	1.63 (0.73–3.58)	0.225
Non-Hispanic Black race-ethnicity	0.98 (0.47–2.05)	0.960
Non-Hispanic other race-ethnicity	2.05 (0.64–5.59)	0.184
Prestroke mRS score, 2–5	0.92 (0.42–1.92)	0.830

LKW indicates last known well; NIHSS, National Institutes of Health Stroke Scale; and OR, odds ratio.

*L1 regularization was used to determine which variables to include in the multivariable models. Age, sex, race-ethnicity, prestroke modified Rankin Scale score, baseline NIHSS score, and LKW-to-door time were forced into the model regardless of L1 regularization results or their univariate significance levels. Age, NIHSS, and LKW-to-door time were analyzed as continuous variables.

†Study arm (mobile stroke unit management vs standard emergency department management) was purposefully omitted from the multivariable model so the findings would generalize to all acute stroke treatment settings.

‡Factors significantly associated with potential missed intravenous thrombolysis opportunities.

While our study demonstrates that withholding IVT in eligible patients with AIS may be a less common practice than it used to be, it also highlights that in those in whom IVT is withheld, major treatment opportunities still exist, as 39% were disabled or dead at 3 months, and 55% had worse function than before their stroke. The PRISMS study (Potential of rtPA for Ischemic Strokes with Mild Symptoms) was a randomized, double-blind clinical trial conducted in the United States comparing intravenous alteplase administered within 3 hours of LKW to oral aspirin in patients with AIS and nondisabling deficits.[18] Among 313 enrolled patients, at 3 months, an excellent functional outcome (mRS score, 0–1) was observed in 78.2% of patients assigned alteplase versus 81.5% of patients assigned aspirin (adjusted risk difference, −1.1% [95% CI, −9.4% to 7.3%]).[18] The ARAMIS study (Antiplatelet vs R-tPA for Acute Mild Ischemic Stroke) was a randomized, open-label, blinded end point assessment, noninferiority trial conducted in China comparing oral aspirin and clopidogrel to intravenous alteplase administered within 4.5 hours of LKW in patients with nondisabling minor AIS.[19] Among 760 randomized patients, a 3-month mRS score of 0 to 1 was observed in 93.8% of patients in the dual antiplatelet group versus 91.4% of patients in the alteplase group (risk difference, 2.3% [95% CI, −1.5% to 6.2%]).[19] Based on the results of these neutral trials, and considering that resolving symptoms and minor deficits were leading reasons for withholding treatment in our analysis, it is uncertain whether IVT would have improved outcomes for eligible patients in BEST-MSU who did not receive IVT; and it is possible that dual

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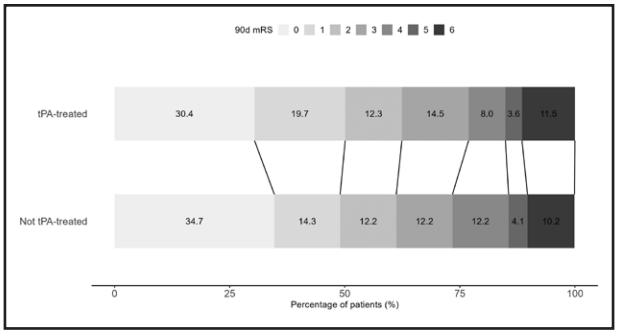

Figure 2. Three-month modified Rankin Scale (mRS) scores for patients according to whether they received intravenous thrombolysis.

Stacked bar graph of the 3-month mRS scores for patients stratified by whether they were or were not treated with intravenous thrombolysis. A score of 0 indicates no symptoms while a score of 6 means the patient has died. Notably, these outcome data are unadjusted, and there were important differences in baseline characteristics between groups. tPA indicates tissue-type plasminogen activator.

antiplatelet therapy or other treatment strategies may confer better odds of good neurological outcomes.[18,19] In fact, in our exploratory analysis comparing propensity score-matched patients treated with IVT to those who were not, 3-month mRS scores were similar between groups. Therefore, more randomized trials are needed to determine the optimal acute treatment strategies for patients with resolving symptoms and minor deficits, particularly those in a 3- to 4.5-hour treatment window.

Our study has several important limitations. First, it was a post hoc secondary analysis of a clinical trial comparing different acute stroke pathways at large centers in the U.S., hence the results should be considered hypothesis-generating and may not generalize to less controlled settings or to centers with fewer resources or experience with stroke. Additionally, the majority of patients were enrolled in Houston, further hampering external validity. Second, despite the large overall sample size of patients with AIS, only 51 potential missed IVT opportunity cases were identified, so our multivariable association analyses may have been underpowered. Third, the study period was 2014 to 2020, and since then AIS treatment patterns may have changed, especially with publication of the ARAMIS trial in 2023, which found dual antiplatelet therapy to be noninferior to IVT in patients with minor nondisabling AIS.[19] Fourth, the higher rate of potential missed IVT opportunities in the ED group versus the MSU group may in part reflect that BEST-MSU trial enrollment depended on a preliminary determination of IVT eligibility, and this was based on patients' condition at the time of presentation to the prehospital paramedic team, not on arrival to the ED. Therefore, patients assigned to the standard management group had a longer interval of observation between enrollment and treatment. Hence, patients who improved while en route to the ED may have been more likely than patients assigned to the MSU group to have IVT withheld because of resolving or minor deficits. Notably, however, many of these patients still did poorly or had worse 3-month functional status than their baseline, suggesting that better treatments may be needed for this subset of AIS patients. Lastly, it is conceivable that family members and other patient contacts were more likely to be at the bedside in an ED than in an MSU setting, and this potential discrepancy could have resulted in additional history being obtained, which affected clinicians' decision-making and their risk-benefit assessment for IVT.

In summary, we found that in modern times, at large centers, 1-in-8 patients presenting to the ED with AIS are not treated with IVT despite lacking contraindications, but that these potential missed IVT opportunities are exceedingly rare aboard MSUs. Patients presenting with milder deficits or later in the 4.5-hour time window are least likely to receive IVT when otherwise eligible, yet at 3 months, many of these patients are disabled or have worse function than before their stroke. Future studies are needed to validate these findings in different settings, including in rural areas and at centers with less stroke expertise. Our findings reinforce the need for constant local and national quality improvement initiatives and medical education to improve acute stroke literacy and management among ED and neurology personnel.

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Supplemental Material

STROBE Checklist

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